

ATTENTION: This utility has been specifically developed for use by municipalities with access to the Watershed Reporting, Adaptive Management and Planning System (WRAMPS) database. Valid login credentials are required for obtaining information stored in WRAMPS for existing best management practices (BMPs) in order to serve as input to this utility.

WRAMPS UTILITY USER DOCUMENT

WRAMPS provides a web-based platform to support Los Angeles County municipalities with municipal separate storm sewer systems (MS4) permit reporting. Since its launch, WRAMPS has become the key data repository of implemented BMPs within Los Angeles County, storing information for over 10,000 BMPs (as of October 2019). Existing BMPs can have a significant impact on downstream hydrology and water quality. To leverage the WRAMPS database in the context of BMP implementation planning, the WRAMPS Utility (Utility) has been developed. WRAMPS can be queried by an authorized user to export a summary file describing the BMP variables and parameters (e.g., BMP type, volume) in the database. The Utility translates this summary file into a model-ready template spreadsheet. BMP variables and parameters for proposed BMPs (as determined by a user) can also be entered into this template spreadsheet. This model approach enables existing and proposed BMPs to be evaluated in the same simulation for comprehensive evaluation of a future condition.

Figure 1 illustrates the relative position of the Utility in the WMMS 2.0 Optimization Framework. As indicated in the figure, the WRAMPS Utility is first in the framework and its output is uploaded to the BMP Opportunities Utility. Reference the *WMMS 2.0 Phase II Report: BMP Model and Optimization Framework* for additional details on the model framework and the suite of utilities.

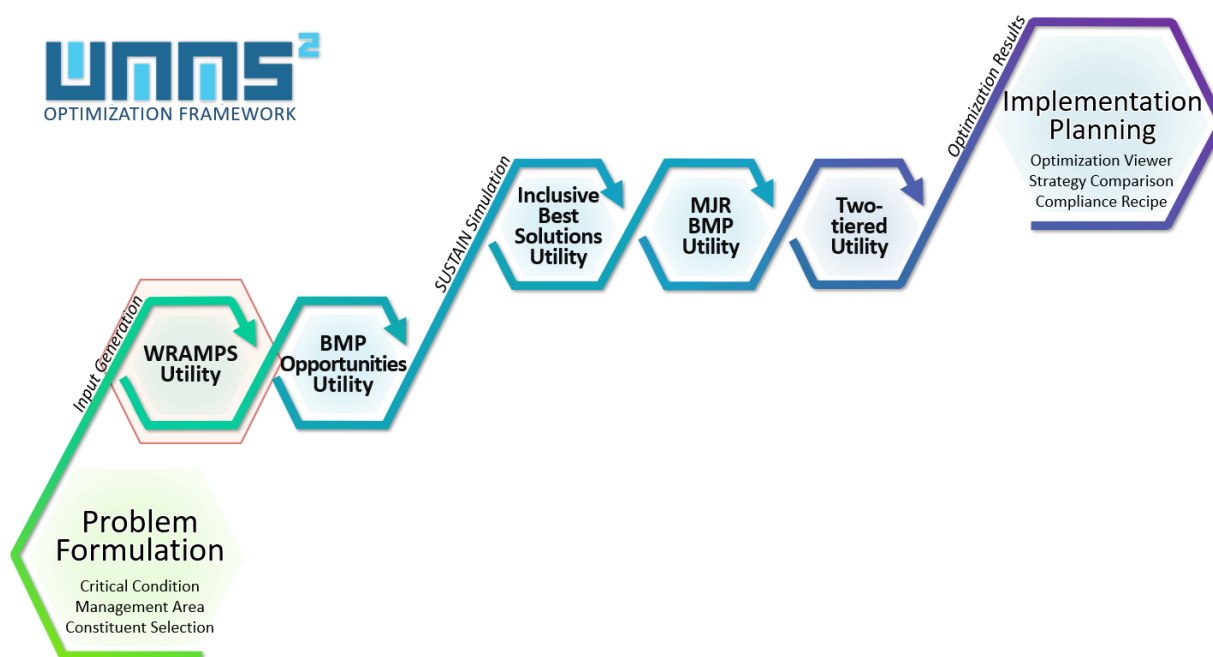


Figure 1. Flow diagram of the WMMS 2.0 Optimization Framework including the WRAMPS Utility.

To support the use of the Utility, this user document has been formulated to describe:

- ▼ The process for obtaining data from the WRAMPS database
- ▼ Characterization of WRAMPS BMPs by the Utility
- ▼ Formatting output and its interpretation into the BMP Opportunities Utility
- ▼ Utility troubleshooting

1 ACCESSING WRAMPS DATA

WRAMPS is a web-based application for MS4 reporting which includes tracking of structural BMPs and estimates of their effectiveness. WRAMPS uses a combination of GIS data, modeling, and web technologies to provide users with visualizations of water capture, data summaries to support MS4 permit reporting, and an inventory of projects within each municipal jurisdiction. Municipalities with access to WRAMPS can use their login credentials to input, modify, and download existing and planned projects in the database. The following steps serve as a guide for users to access and download existing and planned projects for input into the Utility:

Step 1: Go to the [WRAMPS database landing page](#) and click “**Reporting**” to enter user credentials and access the WRAMPS Modules (Figure 2).



Figure 2. WRAMPS log-in page.

Step 2: Navigate to “**Projects Module**” to access the BMP database dashboard (Figure 3).

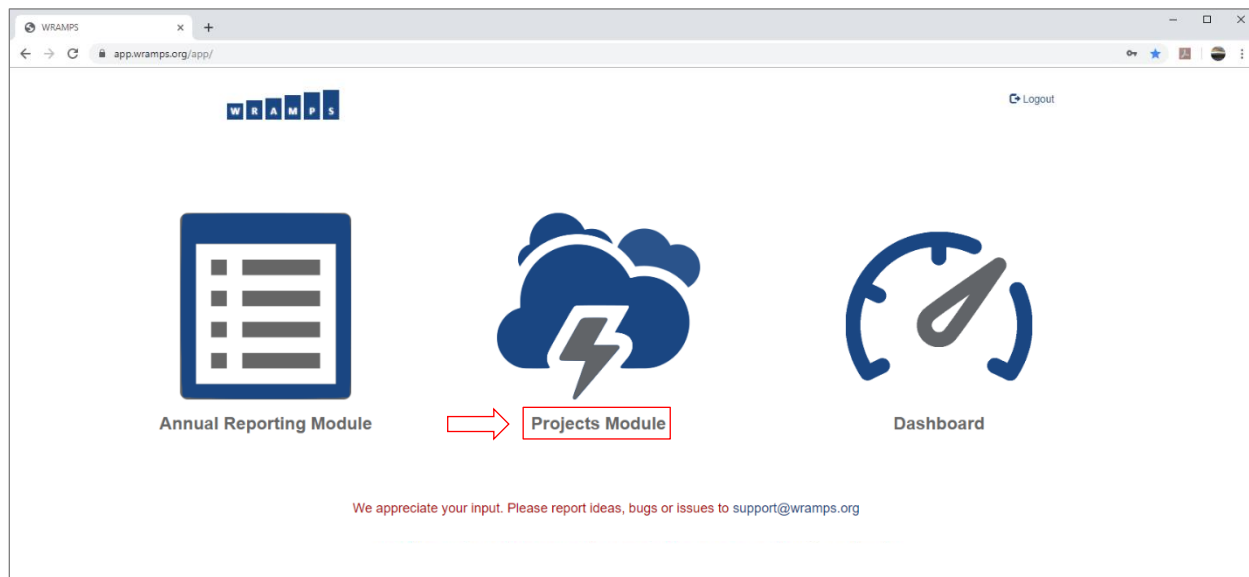


Figure 3. WRAMPS module selection.

Step 3: Once at the primary page for the Project Module database, select “**Dashboard**” at the top of the page to query the BMPs in the database (Figure 4).

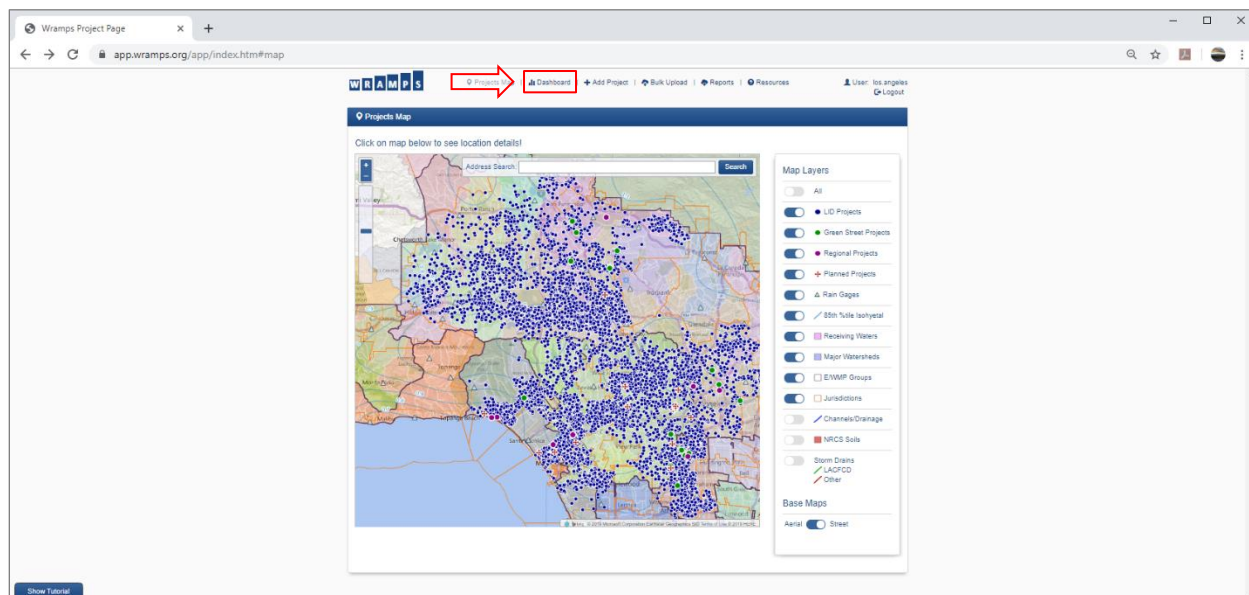


Figure 4. WRAMPS Projects Module.

Step 4: From the dashboard query page, toggle the desired list of projects using the provided search options. For example, select the BMP status (built and/or planned), EWMP Group (e.g., Upper Los Angeles River Group), and jurisdiction (e.g., City of Los Angeles)¹ to populate a list of projects that meet the desired criteria. Select **“Update Dashboard”** to query the specified BMPs (Figure 5). Depending on the number of entries in the selection, this process may take several minutes (typically 1 to 2 minutes, as long as 20 minutes for an especially large download).

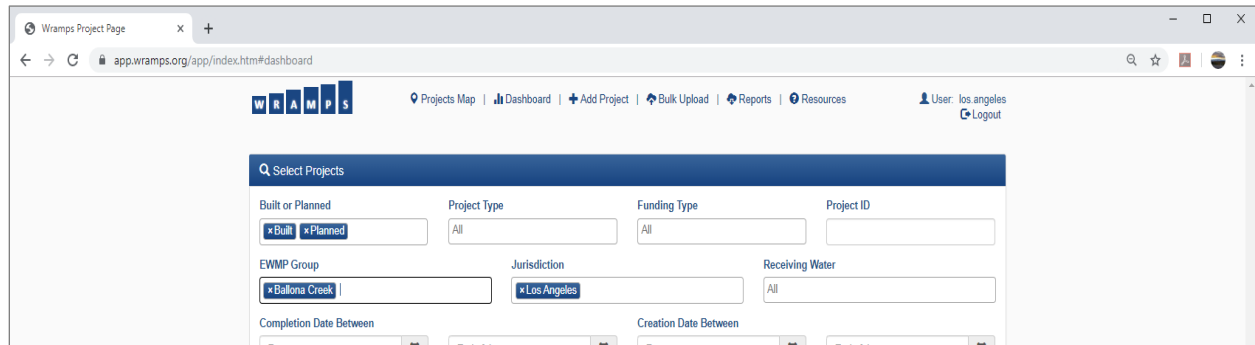
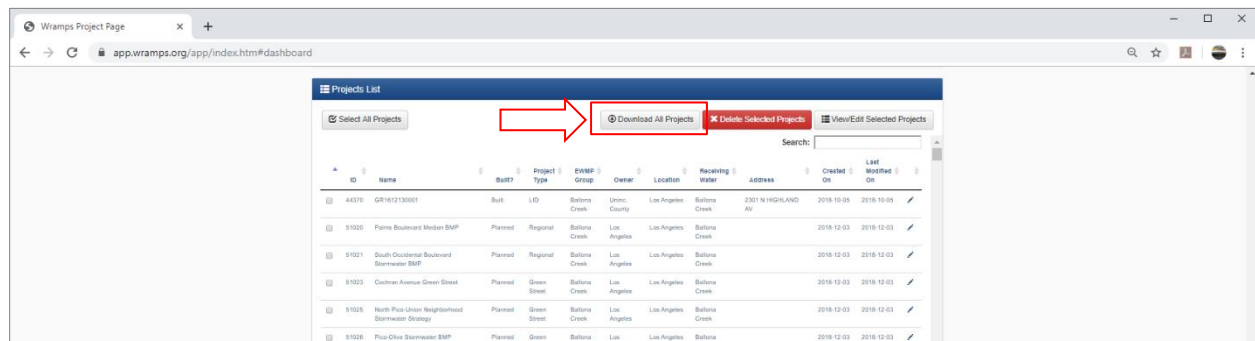


Figure 5. WRAMPS dashboard query page.

Step 5: From the dashboard query page, scroll down to the Project List to view the selection of BMPs that meet the specified criteria. Select **“Download All Projects”** to obtain a bulk download of these BMPs into a spreadsheet (Figure 6). Alternatively, a subset of the queried projects can be selected by using the checkmarks on the left-hand side of the list. Then, **“Download Selected Projects”** can be used to obtain a bulk download of selected BMPs into a spreadsheet. Depending on the number of entries in the selection, this process may take several minutes (typically 1 to 2 minutes, as long as 20 minutes for an especially large downloads). The bulk download will be in the form of a Microsoft Excel spreadsheet (Figure 7).



ID	Name	Built	Project Type	EWMP Group	Owner	Location	Receiving Water	Address	Created On	Last Modified On
46376	GR1812130001	Built	LED	Ballona Creek	Upper County	Los Angeles	Ballona Creek	2001 N HIGHLAND AVE	2018-10-04	2018-10-05
51005	Palos Boulevard Median BMP	Planned	Regional	Ballona Creek	Los Angeles	Los Angeles	Ballona Creek		2018-12-03	2018-12-03
51021	South Occidental Boulevard Stormwater BMP	Planned	Regional	Ballona Creek	Los Angeles	Los Angeles	Ballona Creek		2018-12-03	2018-12-03
51023	Cochran Avenue Green Street	Planned	Green Street	Ballona Creek	Los Angeles	Los Angeles	Ballona Creek		2018-12-03	2018-12-03
51026	North Pico Urban Neighborhood Stormwater Strategy	Planned	Green Street	Ballona Creek	Los Angeles	Los Angeles	Ballona Creek		2018-12-03	2018-12-03
51028	Pico Olive Stormwater BMP	Planned	Green Street	Ballona Creek	Los Angeles	Los Angeles	Ballona Creek		2018-12-03	2018-12-03

Figure 6. WRAMPS BMP Project List.

¹ To access the Excel output from WRAMPS with existing BMP information, please request the file from the jurisdiction or Watershed Management Group lead for the modeling area of interest. Not all jurisdictions use WRAMPS.

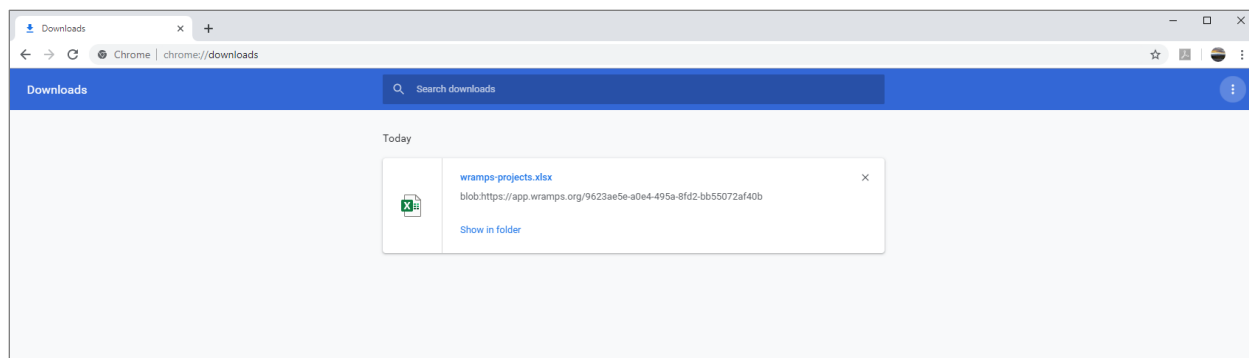


Figure 7. WRAMPS bulk download in a Microsoft Excel spreadsheet.

Step 6: If regional BMP projects (infiltration gallery, large-scale treatment system, or diversion to sanitary sewer) are present within the bulk download, then the extent of their drainage area needs to be entered in the downloaded spreadsheet by the user. Specifically, a user will need to enter whether a regional BMP's drainage area resides within multiple jurisdictions (i.e., a multi-jurisdictional regional [MJR] BMP) or within one jurisdiction (i.e., a non-MJR BMP). A jurisdiction is defined as the intersection between the modeled subwatersheds and municipal boundaries, which allows model results to be summarized across both hydrologic and jurisdictional boundaries. For additional details on jurisdictions and MJR BMPs, reference the *WMMS 2.0 Phase II Report: BMP Model and Optimization Framework*. Figure 8 shows the "Regional HRU Breakdown" tab in the downloaded spreadsheet where the MJR or non-MJR BMP distinction is made. "Yes or No" in column B should be changed by a user to "Yes" if the project is a MJR BMP, or "No" if it is not a MJR BMP.

Column C contains the serialized breakdown of hydrologic response units (HRUs) for each project. Reference Section 2 for how these values are processed by the Utility.

Project Id	MJR BMP?	Serialized HRU Breakdown for WRAMPS Utility in WMMS2
29201	Yes or No	[{"land_use_name":"Road_Primary-All-All-All","time_series_file":"2000_Road_Primary-All-All-All.txt","coverage":5.927810770792265
29202	Yes or No	[{"land_use_name":"Dev_Pervious-D-Med-Confined","time_series_file":"12421_Dev_Pervious-D-Med-Confined.txt","coverage":2.46
29203	Yes or No	[{"land_use_name":"Road_Primary-All-All-All","time_series_file":"2000_Road_Primary-All-All-All.txt","coverage":0.648898401535613

Figure 8. Regional HRU Breakdown tab in the WRAMPS bulk download.

Step 7: Next, the WRAMPS bulk download spreadsheet can be uploaded to the Utility from the WMMS 2.0 web interface (Figure 9).

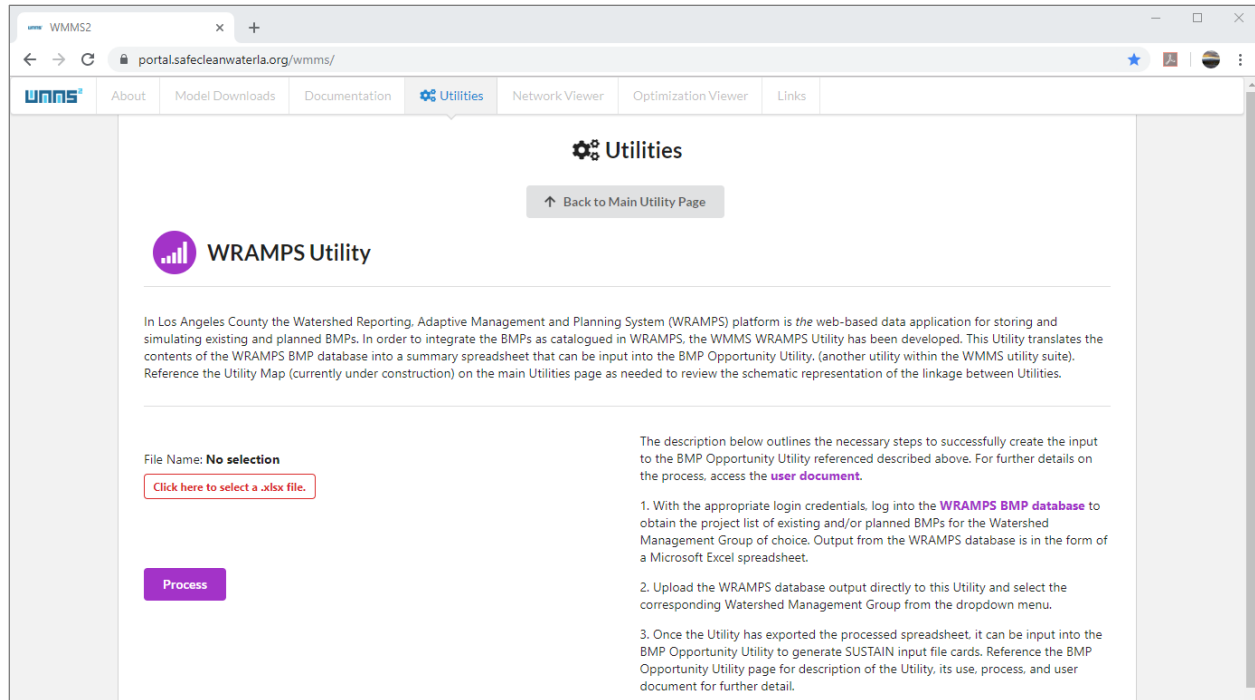


Figure 9. WRAMPS Utility web application page.

2 CHARACTERIZATION OF WRAMPS BMPS

When projects are entered into the WRAMPS database, a user can specify a number of attributes to describe their characteristics (e.g., location, project volume). The WRAMPS bulk download spreadsheet contains all the information recorded at the time of data entry (as well as any subsequent updates). The following set of user-entered attributes are used by the Utility to organize and simulate the BMPs within WRAMPS:

- BMP type (LID, green street, or regional)
- Storage capacity (acre-feet)
- Drainage area (acre)
- Diversion rate, if applicable (cubic feet per second)
- Management group (e.g., Lower Los Angeles River Group)
- Jurished ID (e.g., 100149)

The above listed characteristics are interpreted by the Utility in a specific and intentional approach to model their hydrologic and water quality impact in a SUSTAIN simulation. The following subsections describe how these details are used by the Utility to structure the output spreadsheet.

WRAMPS BMP Location

BMPs in WRAMPS are described by latitude and longitude, enabling their LSPC jurished to be determined. Jurished determination is necessary because SUSTAIN simulations are structured at the jurished-level, meaning all existing/planned BMPs and their model variables (e.g., upstream drainage area) are contained within a single input file (reference the *WMMS2.0 Phase II Report: BMP Model and Optimization Framework* for additional details).

WRAMPS BMP Dimensions

WRAMPS does not currently store the explicit physical dimensions (footprint and depth) of a BMP, which is a required input parameter for SUSTAIN. However, based on the provided project storage capacity and BMP type, which are required input for WRAMPS, an approximation of BMP dimensions can be made. Because BMP depths have been defined in the WMMS 2.0 BMP menu (reference the *WMMS 2.0 Phase II Report: BMP Model and Optimization Framework*), those attributes can be used to assign depths to BMPs in WRAMPS, which are used to back-calculate the associated footprint. This translation between BMPs in WRAMPS to those in the WMMS 2.0 BMP menu is summarized for each BMP in Table 1. This process enables the volumes provided in WRAMPS to be represented in SUSTAIN with explicit dimensions to simulate their impact on hydrology and water quality. If users have specific dimensions for existing and planned BMPs that they prefer, they can overwrite the Utility output with alternative values. **Note:** Plans to update WRAMPS to account for a BMP's specific dimensions when a project is uploaded are in progress. Once completed, the Utility will be able to incorporate these changes to more explicitly represent footprint and depth.

Table 1. Translation of BMPs in WRAMPS to BMP representation in the WRAMPS Utility

WRAMPS Output		WRAMPS Utility Characterization			
BMP Type and Configuration		WMMS BMP Type	Utility ID	Effective Depth ^a	BMP Footprint
LID	Biofiltration	Biofiltration	exLID3	2.75	BMP footprints are back calculated by dividing the storage capacity provided in the WRAMPS database by the assumed effective depth for each BMP (column to the left). $\text{BMP Footprint} = \frac{\text{Volume}}{\text{Effective depth}}$
	Onsite Infiltration	Bioretention	exLID1	2.35	
	Rainfall Harvesting	Bioretention	exLID1	2.35	
Green Street	Green Street (filtration) w/ porous pavement	Biofiltration w/ porous pavement	exLID4 w/ exLID8	2.75 w/ 1.6 ^{b, c}	
	Green Street (infiltration) w/ porous pavement	Bioretention w/ porous pavement	exLID2 w/ exLID6	2.35 w/ 1.6 ^{b, c}	
	Green Street (filtration) w/o porous pavement	Biofiltration	exLID3	2.75	
	Green Street (infiltration) w/o porous pavement	Bioretention	exLID1	2.35	
	Porous pavement	Porous Pavement	exLID5	1.6 ^b	
Regional	Infiltration Facility	Infiltration Facility	Varies ^d	10.2	
	Treatment Facility	Filtration Facility		10	
	Diversion to Sewer	Diversion to Sewer		10	

a – Effective depth. Takes into account ponding height, media depth, void space, etc. See *WMMS 2.0 Phase II Report*.

b – Effective depth for the porous pavement aspect of green street.

c – To model bioretention with porous pavement, routing in the utility output is predefined to ensure these two BMPs are linked (i.e., nested) in SUSTAIN.

d – Regional facilities are given ID's in the order in which they appear in the WRAMPS database (i.e., the first listed regional BMP is "exREG1"). See the "Lumping" subsection below.

WRAMPS BMP Drainage Area

A SUSTAIN simulation requires an explicit drainage area representation of the upstream HRUs. HRUs are discrete elements of land area characteristics (e.g., commercial impervious) used by the LSPC model for simulation of the quantity and water quality of runoff (reference the *WMMS 2.0 Phase I Report: Baseline Hydrology and Water Quality Model* as needed). The following summarizes how HRUs are determined for the three BMPs categories from WRAMPS.

- *Green Streets and LID*: WRAMPS does not contain a detailed HRU drainage area breakdown for Green Streets or LID, only their total area and imperviousness. To generate HRU breakdowns, the WRAMPS Utility proportionally distributes HRUs for the jurished in which the BMP is located, to the drainage area of the BMP. In other words, the overall percent breakdown of HRUs at the jurished-scale are presumed as the relative HRU breakdown at the BMP drainage-scale—the sum of the area will also match the drainage area acreage provided in WRAMPS. Note that if WMMS users have the specific breakdown of HRUs within a BMP drainage area, they can overwrite the default WRAMPS Utility output with their data.
Regional BMPs: WRAMPS has recently been updated to store drainage area delineation and HRU representations for regional BMPs. To be represented in the Utility process, all modeled regional BMPs must have a drainage area (e.g., .shp format) uploaded to the WRAMPS database. When a drainage area is uploaded, the project's HRU breakdown is compiled and exported in the Regional HRU Breakdown tab of the WRAMPS bulk download spreadsheet (see Figure 8, column C). If a drainage area is not uploaded to the WRAMPS database, a user will have to delineate the HRU proportions and enter them manually. See the *BMP Opportunities Utility User Document* for additional details on how this is completed in the BMP Opportunities Utility.
- *MJR BMPs*: Drainage area delineations and HRU breakdowns are not provided for MJR BMPs and are entered manually by the user. See the *BMP Opportunities Utility User Document* for additional details on how this is completed in the BMP Opportunities Utility.

WRAMPS BMP Lumping

Consistent with the BMP Opportunities Utility, WRAMPS LID and green streets (e.g., distributed projects) are modeled in a lumped approach at the jurished scale (reference the *BMP Opportunities Utility User Document* as needed). For example, if there are 50 bioretention cells (e.g., exLID1, reference Table 1) in one jurished, they are characterized and lumped by their cumulative BMP footprint and drainage area in the Utility output. Reference the *BMP Opportunities Utility User Document* as necessary for more details on lumping of distributed projects.

Also consistent with the BMP Opportunities Utility, WRAMPS regional and MJR BMPs, are not modeled in a lumped approach. Instead, regional and MJR BMP projects are simulated explicitly as single facilities which have distinct dimensions, diversion rates, upstream drainage areas, and routing considerations. This representation of regional projects ensures that the most impactful characteristics for determining their performance is maintained to the highest level possible when translated from the WRAMPS database to the Utility. Reference the *BMP Opportunities Utility User Document* as necessary for more details on the representation of regional and MJR projects.

3 UTILITY OUTPUT FORMATTING

Output from the Utility is a Microsoft Excel spreadsheet organized to provide direct input to the BMP Opportunities Utility (Figure 10). In other words, this spreadsheet (the template spreadsheet) can be uploaded directly to the BMP Opportunities Utility to generate input files for a SUSTAIN simulation.

Execution of the BMP Opportunities Utility at this stage would provide a user the ability to assess the efficacy of WRAMPS BMPs. Proposed BMPs (resulting from a user-completed screening/feasibility analysis) can be integrated into the spreadsheet template to be uploaded to the BMP Opportunities Utility. Execution of the Utility in this way enables a user to develop input files that take into account WRAMPS BMPs *and* proposed BMP in the same model simulation for execution in an optimization run (which is facilitated by the next set of SUSTAIN utilities, reference Figure 1). For a complete description of the spreadsheet template, its structure, content, and use, reference the *BMP Opportunities Utility User Document*.

BMP ID and Description				BMP Parameters and Cost									
ID	WMMS BMP ID	Modeled BMP Type	BMP Description	Optimization Viewer Group	Diversion Rate	Weir Height	Soil Depth	POROSITY	FINFILT	Vegetative Parameter	Holtan Growth Index	Underdrain Switch	Underdrain Index
Proposed LID	1 LID1	BIORETENTION	Bioinfiltration	LID, Bioretention	-1	1	3	0.45	5	0.8	1	NoUnderdrain	
	2 LID2	RAINBARREL	Cistern	LID, Other	-1	8	0	0.45	5	0	0	NoUnderdrain	
	3 LID3	POROUSPAVEMENT	Porous Pavement	LID, Other	-1	0.001	3	0.40	5	0	0	NoUnderdrain	
	4 LID4	INFILTRATIONTRENCH	Infiltration Well	LID, Other	-1	62.05	0	0.40	5	0	0	NoUnderdrain	
	5 LID5	DRYPOND	Diversion to Sewer	LID, Other	-1	4	0	0.45	5	0	0	NoUnderdrain	
	6 LID6	DRYPOND	Treatment Unit	LID, Other	-1	4	0	0.45	5	0	0	NoUnderdrain	
Proposed REG	100 REG1	INFILTRATIONTRENCH	Infiltration Facility	Regional Infiltration	-1	10	0.5	0.40	5	0	0	NoUnderdrain	
	101 REG2	DRYPOND	Treatment Facility - Filtration	Regional Treatment	-1	10	0	0.45	5	0	0	NoUnderdrain	
	102 REG3	DRYPOND	Treatment Facility - Sewer Diversion	Regional Treatment	-1	10	0	0.45	5	0	0	NoUnderdrain	
	103 REG4	WETPOND	Treatment Facility - Wetland	Regional Treatment	-1	5	2	0.45	5	0.8	1	NoUnderdrain	
	104 REG5	DRYPOND	Treatment Facility - Detention	Regional Treatment	-1	5	2	0.45	5	0.3	1	NoUnderdrain	
Proposed MJR	999 TBDMP	DRYPOND	To Be Determined	Regional TBD	-1	10	0.5	0.40	5	0	0	NoUnderdrain	
	200 MJRBMP1	INFILTRATIONTRENCH	Infiltration Facility	MJR BMP	-1	10	0.5	0.40	5	0	0	NoUnderdrain	
	201 MJRBMP2	INFILTRATIONTRENCH	Infiltration Facility	MJR BMP	-1	10	0.5	0.40	5	0	0	NoUnderdrain	
	202 MJRBMP3	INFILTRATIONTRENCH	Infiltration Facility	MJR BMP	-1	10	0.5	0.40	5	0	0	NoUnderdrain	
	203 MJRBMP4	INFILTRATIONTRENCH	Infiltration Facility	MJR BMP	-1	10	0.5	0.40	5	0	0	NoUnderdrain	
Existing LID	1000 exLID1	BIORETENTION	Bioinfiltration	Existing LID	-1	1	3	0.45	5	0.8	1	NoUnderdrain	
	1003 exLID2	BIORETENTION	Bioinfiltration (Routed to Porous Pavement)	Existing LID	-1	1	3	0.45	5	0.8	1	NoUnderdrain	
	1004 exLID3	POROUSPAVEMENT	Porous Pavement	Existing LID	-1	0.001	3	0.4	5	0	0	NoUnderdrain	
	1008 exLID4	INFILTRATIONTRENCH	Infiltration Well	Existing LID	-1	62.05	0	0.4	5	0	0	NoUnderdrain	
	1009 exLID5	DRYPOND	Diversion to Sewer	Existing LID	-1	4	0	0.45	5	0	0	NoUnderdrain	
Existing REG	1010 exREG1	BIORETENTION	-	Existing Regional	-1	1	3	0.45	5	0.8	1	NoUnderdrain	
	1011 exREG2	BIORETENTION	-	Existing Regional	-1	1	3	0.45	5	0.8	1	NoUnderdrain	
	1012 exREG3	BIORETENTION	-	Existing Regional	-1	1	3	0.45	5	0.8	1	NoUnderdrain	
	1013 exREG4	BIORETENTION	-	Existing Regional	-1	1	3	0.45	5	0.8	1	NoUnderdrain	
	1014 exREG5	BIORETENTION	-	Existing Regional	-1	1	3	0.45	5	0.8	1	NoUnderdrain	

Figure 10. Output template from the WRAMPS Utility, which is also the input for the BMP Opportunities Utility. Rows labeled with red text are determined by the WRAMP Utility. Items in blue text are determined by a user-completed screening/feasibility analysis.

4 UTILITY TROUBLESHOOTING

A limited number of error messages have been integrated into the Utility to inform users of potential issues for data formatting. As much as possible, errors were written to succinctly and clearly define the issues present so a user could make the appropriate adjustments to their input. The list provided below provides additional context for errors issued on the utility.

Errors

Message: “Worksheet 'LID' does not exist”, “Worksheet 'Regional' does not exist”, or “Worksheet 'Green Street' does not exist”

Context: The incorrect spreadsheet was uploaded to the utility. Ensure that the selected spreadsheet is the intended spreadsheet.

Message: “Jurisheds on WRAMPS input do not match Jurisheds in chosen watershed group”

Context: BMPs within the uploaded document contain jurished IDs that are not consistent with the Watershed Management Group selected from the dropdown menu.

Message: “All uploaded BMP projects need to have a non-zero volume”.

Context: One or more of the listed BMPs in the upload have a non-zero volume. To remedy, the BMP(s) should be updated in the WRAMP’s database download to match the actual project’s volume.

Message: “Network Error”.

Context: Reload the webpage and try again.

Message: “Error parsing Regional HRU Breakdown tab: Error parsing HRU Breakdown tab. Please Enter 'Yes' or 'No' for 'MJR BMP?' column”.

Context: Ensure a “yes” or “no” has been entered in column B of the Regional HRU Breakdown tab for each project.

Message: “SQLite Error 8: ‘attempt to write readonly database’”.

Context: An internal issue has occurred within the Utility. Reload and try again. If the issue persists, email wmms@pw.lacounty.gov for assistance.

Message: “Error parsing Regional HRU Breakdown tab: Error converting value#2 to type 'System.Collections.Generic.List`1[SUSTAINLib.Models.Wramps.HRUInfo]'. Path ", line 1, position 1”.

Context: A change was made to column C of the Regional HRU Breakdown tab. Reupload the shapefiles used to delineate the drainage areas for each selected project in the WRAMPS database. Then, reexport the bulk download spreadsheet. Reupload to the WRAMPS Utility. If the issue persists, email wmms@pw.lacounty.gov for assistance.